

PRACTICAL 7, 8a, 8b:

7.

```
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int*)malloc(sizeof(int));

    printf("Code Segment : %p\n", main);
    printf("Global Segment: %p\n", &global);
    printf("Static Segment: %p\n", &s);
    printf("Heap Segment: %p\n", heap);
    printf("Stack Segment: %p\n", &stack);

    free(heap);
    return 0;
}
```

```
hp@Hima:~/OSSP$ cd OSlab
hp@Hima:~/OSSP$ nano practical7.c
hp@Hima:~/OSSP$ gcc practical7.c -o practical7
hp@Hima:~/OSSP$ ./practical7
Code Segment : 0x5abbf4a2f1a9
Global Segment: 0x5abbf4a32010
Static Segment: 0x5abbf4a32014
Heap Segment: 0x5abc0b190010
Stack Segment: 0x7ffce3f3976c
```

8.

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    a = (int *)malloc(5 * sizeof(int));

    b = (int *)calloc(5, sizeof(int));

    a = (int *)realloc(a, 10 * sizeof(int));

    free(a);
    free(b);

    return 0;
}

```

```

hp@Hima:~/OSSP$ gcc practical8a.c -o practical8a
hp@Hima:~/OSSP$ ./practical8a
hp@Hima:~/OSSP$ nano practical8a.c
hp@Hima:~/OSSP$ gcc practical8a.c -o practical8a

```

8b.

```

GNU nano 2.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        printf("Child Before=%d\n", x);
        x = 500;
        printf("Child After=%d\n", x);
    }
    else
    {
        sleep(2);
        printf("Parent=%d\n", x);
    }

    return 0;
}

```

```
hp@Hima:~/OSSP$ nano practical8a.c
hp@Hima:~/OSSP$ nano practical8b.c
hp@Hima:~/OSSP$ gcc practical8b.c -o practical8b
hp@Hima:~/OSSP$ ./practical8b
Child Before=100
Child After=500
Parent=100
```